

Circuit fault clinic

Teacher guide and worked answers

Use a checked low-voltage circuit to compare electrical conductors and insulators.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

[_passsg/inputs/GROW SOW 2026-27.xlsx](#) | GROW Weekly - Spring | B30:C30

Investigate thermal and electrical conductors and insulators.

Directly develops the stated science outcome. Original secondary-age exemplar activities; diagnostic-led support and stretch follow the SoW. No claim of qualification approval.

Prior learning

Recognise a cell, a lamp and connecting wires.

Know that materials have properties useful for different jobs.

Thermal conductivity and electrical conductivity are different properties; do not treat the thermal test as an electrical result.

Success evidence

Describe a complete conducting loop through the cell, lamp and sample.

Use a known-conductor check before interpreting an unlit lamp.

Choose conductor and insulation materials for a lead, using test evidence.

Equipment

Paper route: common evidence page, selected pupil response route and pencils.

Optional practical per pair: one 1.5 V AA cell in a school holder, matching 1.5 V lamp and holder, switch, four insulated crocodile-clip leads.

Safe prepared dry samples: bare copper strip, aluminium foil strip with folded edges, dry card, rigid plastic strip; optional prepared coated-metal sample.

Preparation

Teacher checks complete kit, cell condition, matching lamp rating and all leads before the lesson. Use the school-approved low-voltage kit and local procedure.

Prepare a known copper test strip with good contact surfaces. Check it immediately before and after the sample sequence.

Prepare samples and pupil copies in advance. Use the full paper route if any practical kit is unavailable. Do not substitute loose LEDs without the correct circuit.

Practical care

Use only the single 1.5 V cell and matching school lamp circuit described; never use mains sockets, plugs or electrical appliances.

Keep the lamp in series. Never join cell terminals directly with wire or a metal sample; this can heat the cell and leads.

Switch off before changing samples. If any component becomes warm, disconnect safely and tell the teacher.

Use dry prepared solid samples only. Do not test bodies, liquids or unknown objects; do not dismantle cables or scrape coatings.

Keep components out of mouths; teacher manages damaged cells and sharp or broken materials.

Ready to teach alternative

The complete paper fault-clinic uses supplied constructed observations and circuit diagrams. It assesses interpretation and design without handling electrical equipment.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	You are the circuit fault team
2	Retrieval	2-5	Find the missing idea
3	I do	5-13	Trace the whole loop
4	I do	Within stage	One lead needs two properties
5	I do	Within stage	Do not trust a single dark lamp
6	We do	13-21	Fault clinic: sort the case cards
7	We do	Within stage	Repair the reasoning, not the sample
8	We do	Within stage	Build the class evidence table
9	Hinge	21-24	The control fails. What now?
10	You do	24-34	Make your diagnostic report
11	You do	Within stage	If using the optional kit
12	You do	Within stage	Use a scientific conclusion
13	Review	34-38	Compare diagnoses
14	Review	Within stage	Use our voice to improve the test
15	Exit	38-40	Exit diagnosis

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening You are the circuit fault team

Explain the repair-clinic brief. Success is a defensible diagnosis, not fastest construction. Establish that no real appliances, mains equipment, bodies or liquids are tested.

2 Retrieval Find the missing idea

R1 point to components in the next diagram if names are unfamiliar. R2 no: the loop must connect both cell terminals through the lamp. R3 no: thermal concerns energy transfer due to temperature difference; electrical concerns current. This lesson investigates electrical behaviour.

3 I do Trace the whole loop

8 minutes shared across three I do slides. Trace from one cell terminal, through wire, sample, lamp and closed switch to the other terminal. This is one cell, not a mains supply. The diagram has a switch shown closed. Teacher demonstrates known copper first; with valid kit it lights. Current is not used up by the lamp; electrical energy is transferred there.

4 I do One lead needs two properties

Shared I do time. Think aloud: I need charge to move along the inside, so copper is useful. I want to reduce unwanted contact with that conducting path, so suitable insulation covers it. The enlarged cutaway is explanatory only. A classroom result does not certify insulation for real wiring.

5 I do Do not trust a single dark lamp

Shared I do time. Show the contact diagram. If clips touch paint, the test has not reached the metal underneath. Do not ask pupils to scrape it. If either copper check fails, stop: repair the circuit before classifying anything. Lamp brightness is an indicator with a detection limit, not proof of exactly zero current.

6 We do Fault clinic: sort the case cards

8 minutes shared across the We do slides. Pair roles: evidence reader and diagnostic checker; one person can work silently. Cases A copper lit, B dry plastic dark with controls pass, C foil dark but control fails, D painted metal dark with clips on paint, E bare aluminium lights. Expected categories: A/E conductor; B poor conductor or insulator in this tested circuit; C/D uncertain.

7 We do Repair the reasoning, not the sample

Shared We do time. C: disconnect and teacher checks cell, lamp, leads and switch, then repeat known conductor and sample. D: use a safe prepared sample with proper contact to the intended material, or mark uncertain. Do not scrape, cut, add cells or bypass the lamp. A fault clinic values the answer not enough evidence.

Reveal: C: check the circuit. D: check which material the clips actually touch.

8 We do Build the class evidence table

Shared We do time. All samples dry, contacts valid, same kit; these are constructed observations. Ask which two materials provide conductor evidence and which two support poor-conductor classification. The lamp does not rank copper versus aluminium quantitatively.

9 Hinge The control fails. What now?

Collect an independent choice. If A or C selected, point to the failed known conductor. If D selected, remind pupils never to raise voltage or bypass checks. Repair logic before You do.

A. Plastic is definitely the cause. - No. The copper control also failed, so the circuit may be faulty.

B. Check the circuit, then repeat the tests. - Yes. A passing control is needed before interpreting the sample result.

C. Copper has become an insulator. - No. A faulty lamp, cell, lead or contact is a more relevant explanation to check.

D. Add many more cells to force a result. - No. Keep to the approved low-voltage kit and ask the teacher to check it.

10 You do Make your diagnostic report

10 minutes shared over three You do slides. Optional pairs may test four prepared materials within this interval: teacher supplies checked loop, then pupils change one sample with switch off, observe and record. Paper route is complete without the practical. Actual observations may differ: record them honestly and diagnose, do not overwrite them with the supplied examples.

11 You do If using the optional kit

Shared You do time. A 10-minute test is possible only with preassembled checked kit and prepared samples. Any delay: use the provided observation cards immediately. Do not add curriculum content or compromise the review to chase a working bulb.

12 You do Use a scientific conclusion

Shared You do time. Ask for evidence and reason. Supported pupils may dictate. Stretch note: graphite with a very small meter-detected current may fail to visibly light this lamp; zero visible light is not the same as zero current. No meter practical required.

13 Review Compare diagnoses

4 minutes shared across two Review slides. Expected response: a failed control means the setup cannot yet diagnose the sample. Copper inside and plastic outside meet different functions. Accept aluminium inside if evidence linked; avoid claims about suitability for all real cable manufacturing.

14 Review Use our voice to improve the test

Light Lundy loop: pupils can point, write or speak. Teacher notes the chosen improvement and explains how the next investigation will use it. Do not make a popularity vote determine a scientific fact.

15 Exit Exit diagnosis

E1 current needs a connected conducting path through this loop. E2 confirms kit works throughout; failed checks invalidate the classification. E3 copper/aluminium inside, appropriate plastic insulation outside, with jobs explained. Ask one transfer question next lesson if response is only a material list.

Answers and assessment

R1-R3

R1 cell provides the electrical supply, lamp is the light-emitting component, wires make connections. R2 no: there must be a complete loop through both cell terminals and the lamp. R3 no: thermal and electrical conductivity are distinct properties, investigated differently.

W1 / cases A-E

A and E give conductor evidence: bare copper/aluminium light the lamp with controls and contacts valid. B supports classification as a poor conductor/insulator under this low-voltage test: dry plastic gives no visible glow with passing controls. C is uncertain because the known-conductor control fails. D is uncertain for the metal core because clips touch coating; the test concerns the contacted surface.

W2

C: teacher disconnects, checks cell, lamp, leads and switch, restores passing copper check, then repeats the sample and control. D: teacher supplies a safe prepared sample with sound contact to exposed intended material, or records uncertain. Do not scrape/cut samples, add cells or bypass the lamp.

We do table

Copper and bare aluminium provide conductor evidence; dry card and rigid plastic are poor conductors/insulators in this tested setup. No visible glow does not demonstrate zero current or electrical safety at other voltages. These observations are not a quantitative ranking of metals.

Hinge A-D

A wrong: the failed copper check means the plastic cannot yet be blamed. B correct: check circuit and repeat with passing controls. C wrong: investigate equipment/contact rather than claiming copper became an insulator. D wrong: do not add cells; keep the stated safe kit and refer to teacher.

S1-S2

S1 cell, lamp and sample joined in a loop through both cell terminals; an air gap interrupts it and prevents a visible lamp current in this kit. S2 A lights/conductor; B no glow/insulator evidence in this test; C no glow/uncertain because control failed.

S3-S5

S3 copper control; circuit. S4 copper inner path because charge can move readily; plastic outer cover because it resists charge movement and reduces unwanted contact. S5 only that this lamp did not visibly light; very small current may be below the lamp indication threshold.

N1

Copper A: lit, conductor, valid controls. Plastic B: no visible glow, insulator evidence in this test, valid controls and contacts. Foil C: no visible glow, uncertain, failed control. Coated metal D: no glow, uncertain for metal, clips touch paint. Aluminium E: lit, conductor, valid controls.

N2-N3

N2 with switch off use known copper and then switch on to confirm kit works; switch off, connect the prepared dry sample with sound contact and observe; switch off and restore copper, then verify the control still works. Accept equivalent sequence including no bypass of lamp. N3 diagram labels conducting inner copper/aluminium and insulating outer plastic; explain current path and reduced unwanted contact. No real cable dismantling.

N4-N5

N4 low-voltage no-glow results do not establish no current under all conditions. Higher voltages and different conditions can be hazardous; never test mains equipment or bodies. N5 no: visible lamp observations are qualitative, not a measured conductivity ratio. Quantitative comparison needs suitable measurements and controlled sample dimensions and conditions.

T1-T2

T1 reject only metals: graphite is a non-metal and the meter detects a current. T2 the small current may be below that needed for visible lamp glow. A dark lamp reports no visible light in this circuit, not mathematically zero current. Do not increase class voltage to seek a response.

T3-T4

T3 same cell, lamp, leads, sample length/contact spacing and safe contact quality; change sample, check known copper before and after, switch off for changes, record observations and repeat. Limitation: lamp threshold and sample geometry mean qualitative evidence, not exact intrinsic conductivity. T4 C has failed equipment/control validity; D lacks valid contact with the intended metal core.

T5

Conducting inner copper or aluminium enables the current path; insulating outer material helps isolate it. These class observations cannot certify a product for mains wiring, durability or all voltages, nor prove zero leakage current.

E1-E4 / optional record

E1 a connected conducting path is needed around this circuit for a lamp current. E2 establishes the kit worked immediately before and after sample testing; failures require rechecking. E3 copper/aluminium conducting interior; suitable plastic insulating exterior, with functions. E4 relevant improvements: check contacts, repeat with controls, separate observation and inference. Optional actual-test record has no fixed observations: assess truthful recording and sound inference, including uncertain when controls fail. Reflection accepts any evidence-linked revision.

Assessment / re-teach

Secure objective: can trace complete loop, interprets a dark result only after passing controls and valid contacts, and justifies conductor/insulator functions. A material sort without controls is incomplete. Re-teach with Case C beside A: same known copper, different lamp response. Ask what changed about the evidence before retesting conceptual understanding.

Teaching and assessment notes

40-minute sequence: 2 opening + 3 retrieval + 8 I do + 8 We do + 3 hinge + 10 You do + 4 review + 2 exit. Zero-minute slides share the stage allocation.

The lesson is complete on paper. Optional kit work requires prepared checked apparatus and teacher supervision, and should not displace independent reasoning or exit evidence.

Keep thermal and electrical meanings separate. Some materials are good conductors of both, but this does not make the two processes identical.

Circuit drawing is a simplified representation with exaggerated component sizes; charge carriers and fields are not visible. The cable is a model cutaway, never a prompt to strip wire.

Treat classifications as useful relative descriptions under given conditions. No lamp glow is evidence of poor conduction only once contacts and known-conductor checks pass. No result here certifies electrical safety.

Access and responsive teaching

Supported

Use page 2: trace the loop, classify three provided observations and complete one cable sentence. Read or scribe as required.

Standard

Use page 3: interpret the whole result table, diagnose a failed control and justify both cable layers.

Stretch

Use the Stretch route: evaluate lamp sensitivity and a graphite counterexample, then qualify the electrical-property claim. Same objective, more demanding evidence.

Regulation

Pairs may choose checker, recorder or observer roles. Provide a seated paper route and quiet independent response. Never require touching or handling for participation; no timed wiring contest.

Misconceptions to check

An unlit lamp proves the sample is an insulator.

A faulty circuit or poor contact can also prevent light. Check a known conductor before and after the test. Check: Present the failed copper-control card; ask whether the sample can be classified.

The lamp uses up the current.

In this steady series loop the current is the same at each point. Energy is transferred by the lamp; charge is not used up.

Check: Ask whether a second wire is needed to complete the loop.

All non-metals are insulators.

Many common dry non-metals are poor conductors, but graphite is a conducting non-metal.

Check: Use the supplied stretch meter evidence; never increase voltage to make a class sample conduct.

A material is safe to touch at any voltage if the little lamp does not light.

This low-voltage result is not a general safety test. A small or undetected current can still exist; conditions and voltage matter.

Check: Ask whether school test results permit touching household wires. They do not.

Word	Meaning
electrical conductor	A material through which electric charge can move readily.
electrical insulator	A material through which electric charge moves much less readily.
current	The rate of flow of electric charge.
complete circuit	A connected loop through which a current can flow.
control check	A known test used to check that the method or equipment works.

Scientific references

[Institute of Physics: Conductors and insulators](#)

Checked 9 September 2026. Conductivity is not an absolute two-category boundary; low-voltage lamp results must not imply human safety.

[Institute of Physics: Electric circuits and fields guidance notes](#)

Checked 9 September 2026. Teacher equipment checks, simple series-circuit troubleshooting and distinction between current and transferred energy. Use modern school kit guidance, not historical battery advice.

[National Grid: Conductors and insulators education resource](#)

Checked 9 September 2026. Supports a low-voltage material-gap test and graphite as a conducting non-metal. This original lesson excludes hazardous or unsuitable listed materials.